

The Central Limit Theorem

- Let $X_1, X_2, \dots, X_n$ denote the observations of a random sample from a distribution that has mean μ and positive variance σ^2 . The the random variable

$$Y_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sqrt{n}\sigma} = \sqrt{n} \frac{\bar{X} - \mu}{\sigma}$$

converges in distribution to standard normal distribution $N(0, 1)$.

- In general, $n > 30$,

$$\sum_{i=1}^n X_i \sim N(n\mu, n\sigma^2)$$

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

- Let $X_1, X_2, \dots, X_n$ denote a random sample from $b(1, p)$. Let $Y_n = X_1 + X_2 + \dots + X_n$ which is $b(n, p)$. When we do not use poisson approximation, we can use the fact that

$$\frac{Y_n - np}{\sqrt{np(1-p)}} = \sqrt{n} \frac{\bar{X} - p}{\sqrt{p(1-p)}}$$

has a limiting distribution $N(0, 1)$.

- Continuity correction: $n = 100$ and $p = 1/2$,

$$\mathbb{P}(Y = 48, 49, \dots, 52) = \mathbb{P}(47.5 < Y < 52.5)$$

- Let $\bar{X}$ be the sample mean of a random sample of size 75 from uniform(0, 1). $\mu = 0.5$ and $\sigma^2 = 1/12$.
 - The pdf of $\bar{X}$ is very complicated, which is the arcs of 75 different polynomials of degree 74.
 - By the CLT,

$$\begin{aligned}\mathbb{P}(0.45 < \bar{X} < 0.55) &= \mathbb{P}\left(\frac{0.45 - 0.5}{\sqrt{1/12}/\sqrt{75}} < \frac{\bar{X} - 0.5}{\sqrt{1/12}/\sqrt{75}} < \frac{0.55 - 0.5}{\sqrt{1/12}/\sqrt{75}}\right) \\ &\approx \mathbb{P}(-1.5 < Z < 1.5) = 0.866\end{aligned}$$

References

- [1] *Introduction to Mathematical Statistics* by Hogg, McKean, Craig
- [2] *Probability and Statistics: The Science of Uncertainty* by Evans and Rosenthal
- [3] *A First Course in Probability* by Ross
- [4] *John E. Freund's Mathematical Statistics with Applications* by Irwin Miller Marylees Miller